

Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

- Answer Choices:
- A. 2.48×10^{-8} cm
 - B. 5.92×10^7 cm/sec
 - C. 8.66×10^{-6} cm
 - D. 3.14×10^{-5} cm
 - E. 9.81×10^{-4} cm
 - F. 5.27×10^{-9} m
 - G. 1.23×10^{-7} cm
 - H. 4.14×10^{-10} m
 - I. 6.625×10^{-27} erg sec
 - J. 0.512×10^6 eV

Reasoning + Answer:

<think>

- First , convert 1 eV to joules . $1\text{ eV} = 1.602 \times 10^{-19}\text{ J}$
- The de Broglie wavelength is given by $\lambda = h / p$, where h is Planck's constant and p is the momentum .
- Momentum can be calculated as $p = \sqrt{2 * m * E}$ for non-relativistic speeds , where m is the mass of the electron and E is the kinetic energy .
- The mass of an electron is approximately $9.109 \times 10^{-31}\text{ kg}$.
- Use these values to find the wavelength .
- Planck's constant $h = 6.626 \times 10^{-34}\text{ Js}$
- We first find the momentum p and then use the de Broglie equation to find λ .

$$\lambda = h / p$$
$$p = \sqrt{2 * m * E}$$
$$E = 1.602 \times 10^{-19}\text{ J} , m = 9.109 \times 10^{-31}\text{ kg}$$

After calculating , the wavelength λ will be around $1.23 \times 10^{-7}\text{ cm}$.

</think>

<answer> $1.23 \times 10^{-7}\text{ cm}$ </answer>